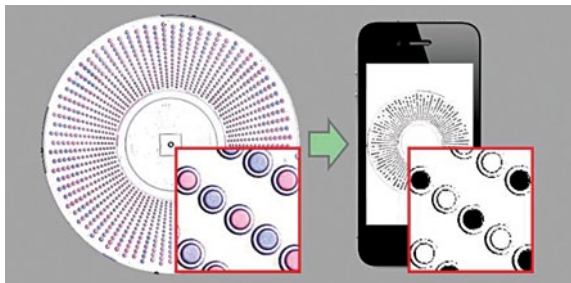


vide unambiguous quantification of single nucleic-acid molecules that can be performed by any mobile phone camera.

The work uses a micro-fluidic technology called SlipChip, which was invented in Ismagilov Lab many years ago. SlipChip serves as a portable lab-on-a-chip and can be used to quantify concentrations of single molecules. Each



Wells turn blue if a particular nucleic acid molecule is present. A mobile phone image of the wells can then undergo ratiometric spectral processing to provide a quantification of target molecules (Image courtesy: Ismagilov Lab)

SlipChip encodes a complex program for isolating single molecules (such as DNA or RNA) along with chemical reactants in nanoliter-size wells.

The program also controls complex reactions in each well; the chip consists of two plates that move, or slip, relative to one another, with each slip joining or separating the hundreds or even thousands of tiny wells, either bringing reactants and molecules into contact or isolating these. The architecture of the chip enables users to have complete control over these chemical reactions and can prevent contamination, making it an ideal platform for a user-friendly, robust diagnostic device.

Consumer drone that can fly itself

DJI Phantom 4 is a consumer drone that can fly itself, avoids obstacles and follows people. It uses computer vision that autonomously avoids obstacles. This quadcopter is equipped with five cameras; two on the front, two on the belly and one primary 4K camera. All images from these



Phantom 4

cameras are run through a computer vision software that processes these and 3D-maps the environment. Regardless of the mode the drone is flying in, it senses obstacles and either ascends to fly over it or comes to a screeching halt instantly and hovers. The crash-avoidance feature can be turned off for expert flyers to get closer to obstacles.

DJI have used their own camera on Phantom 4, and it is capable of making hi-res 4K videos, saved on a microSD card. Viewport of the primary camera is 60°, with a 30° vertical view. Phantom 4 empowers even the most novice of flyers to fly the drone and shoot amazing pictures, without using sticks at all.

Tiny robots pull a two-ton car

A fleet of teensy robots, collectively weighing less than an apple, moved a 1800-kilogram car and driver. The secret behind these tiny-but-mighty bots is an adhesive inspired by gecko feet.

According to David Christensen, a mechanical engineering doctoral candidate at Stanford University in California, USA, who helped design the robots, a synthetic gecko adhesive was used that is turned on when shear force is applied and then turned off as soon as it released. The robots lock onto the surface when they want to but are never actually stuck.

The tiny bots are made using a tiny battery that powers the motor, which lifts a metal arm anchored to a towing cable. At the bottom of each bot's feet is an adhesive that makes the robot parts operate like a gecko's foot. When pulled vertically, the adhesive offers no resistance, but when tugged sideways, parallel across a surface, the material strongly resists motion.

Samsung's VR headphones trick your brain into feeling motion

Entrim 4D headset from Samsung makes virtual reality (VR) way more realistic. The new headphones trick your brain into feeling like you are moving, even when you are standing completely still. Entrim, which works with Samsung's Gear VR headset, sends motion signals to your brain via your inner ear, so you feel like you are moving with the images you are experiencing in VR. These do not provide vibrations or any type of physical feedback, so they do not physically feel different than any other type of headphones.

Using a combination of algorithms and galvanic vestibular stimulation, a safe and simple technique that sends specific electric messages to a nerve in the ear, the VR accessory synchronises your body with changing movements in video content. Electrical signals, like the ones used to help restore balance in stroke patients, are delivered via headphones equipped with electrodes that correspond with movement data input by engineers.